

Entropic Portfolio Optimization: a Disciplined Convex Programming Framework

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[First version: January 2021] [Latest version: April 2021]

Abstract

This work presents a disciplined convex programming framework for entropic value at risk (EVaR) based on exponential cone programming. This framework allows us to use EVaR in several convex portfolio optimization problems like maximize the EVaR adjusted return, constraints on EVaR or risk parity EVaR. Also, we propose a new portfolio optimization framework based on an extension of EVaR but applied to drawdowns distribution that we called entropic drawdown at risk (EDaR). Then, we run some numerical examples of EVaR and EDaR portfolio optimization frameworks using Python, Riskfolio-Lib package and MOSEK solver. Finally, we test the efficiency of EVaR and EDaR frameworks in large scale problems respect to conditional value at risk (CVaR) and conditional drawdown at risk (CDaR).

Keywords: risk analysis, portfolio optimization, coherent risk measures; convex optimization, portfolio selection; entropic value at risk; risk parity; exponential cone programming.

JEL Codes: C61, G11

Acknowledgement: We thanks to PhD Amir Ahmadi-Javid (Amirkabir University of Technology) for comments on the manuscript.

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1 Introduction

Convex programming is a subclass of nonlinear programming (NLP) that unifies and generalizes least squares (LS), linear programming (LP), convex quadratic programming (QP) and cone programming (CP). Disciplined convex programming, defined in [Liberti \(2006\)](#), is a modeling methodology that imposes a set of conventions that one must follow when constructing convex programs. The advantage of this methodology is that the transformations necessary to convert disciplined convex programs into solvable form can be fully automated. There are several optimization packages that supports disciplined convex programs, for example CVXPY, CVX, MOSEK, among other packages.

The entropic value at risk (EVaR) is a new coherent risk measure proposed by [Ahmadi-Javid \(2012\)](#). It is the upper bound of Chernoff inequality for the value at risk (VaR) and conditional value at risk (CVaR). Also [Ahmadi-Javid and Fallah-Tafti \(2019\)](#) show that EVaR can be used in portfolio optimization for large scales problems. However, the framework proposed in [Ahmadi-Javid and Fallah-Tafti \(2019\)](#) is not disciplined convex programming, so it cannot be solved by the softwares mentioned above. This is a problem if we want to implement optimization problems like maximize the EVaR adjusted return, risk parity EVaR or constraints on EVaR.

This work proposes a framework based on exponential cone programming that solves the problems mentioned above. We also extend the concept of EVaR to drawdowns creating the entropic drawdown at risk (EDaR) portfolio optimization framework. First, we show how to express the EVaR as the sum of a convex function and an affine function. Then, we express this convex function as a disciplined convex programming expression using variables and exponential cone programming. Then, we propose several portfolio optimization problems using the disciplined convex framework of EVaR and EDaR. Finally, we make numerical examples of EVaR and EDaR portfolio optimization using Python, Riskfolio-Lib package and MOSEK solver.

2 Portfolio Optimization Frameworks

2.1 Mean-Risk Portfolio Optimization

The mean-risk optimization is a generalization of the model proposed by [Markowitz \(1952\)](#), that can be applied to any convex risk measure. Mean-risk optimization begins

selecting $N \in \mathbb{N}$ assets, a series of historical returns $r \in \mathbb{R}^{T \times N}$ where $T \in \mathbb{N}$ is the number of observations, $\mu \in \mathbb{R}^N$ is the vector of expected returns of r , $x \in \mathbb{R}^N$ is the vector of weights or portfolio, μx^τ ¹ is the expected return of portfolio x , $\phi(x)$ is the expected risk of portfolio x and rx^τ is series of historical portfolio returns. The mean-risk optimization can be resumed as four convex optimization problems:

- Maximization of the expected return given a level of risk $\bar{\phi}$.

$$\begin{aligned}
 & \max_x \quad \mu x^\tau \\
 & \text{s.t.} \quad \phi(x) \leq \bar{\phi} \\
 & \quad \sum_{i=1}^N x_i = 1 \\
 & \quad x_i \geq 0 ; \forall i = 1, \dots, N
 \end{aligned} \tag{1}$$

- Minimization of the expected risk given a level of expected return $\bar{\mu}$.

$$\begin{aligned}
 & \min_x \quad \phi(x) \\
 & \text{s.t.} \quad \mu x^\tau \geq \bar{\mu} \\
 & \quad \sum_{i=1}^N x_i = 1 \\
 & \quad x_i \geq 0 ; \forall i = 1, \dots, N
 \end{aligned} \tag{2}$$

- Maximization of a risk averse utility function given a risk aversion parameter $\lambda \in [0, \infty)$.

$$\begin{aligned}
 & \max_x \quad \mu x^\tau - \lambda \phi(x) \\
 & \text{s.t.} \quad \sum_{i=1}^N x_i = 1 \\
 & \quad x_i \geq 0 ; \forall i = 1, \dots, N
 \end{aligned} \tag{3}$$

- Maximization of the risk adjusted return, based on the idea of [Sharpe \(1964\)](#), assuming that the return of the risk free asset is r_f . This problem can be posed

¹The operator $*^\tau$ is the operator transpose of a matrix.

as the following form:

$$\begin{aligned}
& \max_x \quad \frac{\mu x^\tau - r_f}{\phi(x)} \\
& \text{s.t.} \quad \sum_{i=1}^N x_i = 1 \\
& \quad \quad x_i \geq 0 ; \forall i = 1, \dots, N
\end{aligned} \tag{4}$$

This problem is quasi convex, so it cannot be solved with a convex programming software; however, it can be transformed into a convex programming problem using the transformation of [Charnes and Cooper \(1962\)](#) as long as it is assumed that $\mu x^\tau - r_f > 0$. After the transformation the problem takes the following form:

$$\begin{aligned}
& \min_{y, k} \quad \phi(y) \\
& \text{s.t.} \quad \mu y^\tau - r_f k = 1 \\
& \quad \quad \sum_{i=1}^N y_i = k \\
& \quad \quad k \geq 0 \\
& \quad \quad y_i \geq 0 ; \forall i = 1, \dots, N
\end{aligned} \tag{5}$$

where the optimal portfolio is obtained making the transformation $x = y/k$.

2.2 Risk Parity Portfolio Optimization

The general vanilla risk parity portfolio formulated by [Bruder and Roncalli \(2012\)](#) has the following form:

$$\begin{aligned}
& \min_y \quad \phi(y) \\
& \quad \quad \sum_{i=1}^N b_i \ln(y_i) \geq c \\
& \quad \quad \sum_{i=1}^N b_i = 1 \\
& \quad \quad y_i \geq 0 ; \forall i = 1, \dots, N
\end{aligned} \tag{6}$$

Where b is a vector of risk concentration constraints of size $1 \times N$ and c is an arbitrary constant. The optimal portfolio is obtained making the transformation $x = y / \sum_{i=1}^N y_i$.

2.3 Entropic Value at Risk (EVaR)

The EVaR is the upper bound of value at risk (VaR) and conditional value at risk (CVaR) obtained from Chernoff inequality. Formally is defined as:

$$\text{EVaR}_\alpha(X) = \min_{z>0} \left\{ z \ln \left(\frac{1}{\alpha} M_X \left(\frac{1}{z} \right) \right) \right\} \quad (7)$$

where $M_X(t)$ is the moment generating function of t and α the significance level. In a sample version the EVaR can be expressed as:

$$\text{EVaR}_\alpha(X) = \min_{z>0} \left\{ z \ln \left(\frac{1}{\alpha} \sum_{j=1}^T p_j \exp \left(\frac{X_j}{z} \right) \right) \right\} \quad (8)$$

where p_j is the probability of scenario j . If we assume that each scenario has the same probability, then $p_j = \frac{1}{T}$, and the EVaR can be expressed as follow:

$$\begin{aligned} \text{EVaR}_\alpha(X) &= \min_{z>0} \left\{ z \ln \left(\frac{1}{\alpha} \sum_{j=1}^T \frac{1}{T} \exp \left(\frac{X_j}{z} \right) \right) \right\} \\ \text{EVaR}_\alpha(X) &= \min_{z>0} \left\{ z \ln \left(\frac{1}{\alpha T} \sum_{j=1}^T \exp \left(\frac{X_j}{z} \right) \right) \right\} \\ \text{EVaR}_\alpha(X) &= \min_{z>0} \left\{ z \ln \left(\sum_{j=1}^T \exp \left(\frac{X_j}{z} \right) \right) + z \ln \left(\frac{1}{\alpha T} \right) \right\} \end{aligned} \quad (9)$$

We can see that the expression $z \ln \left(\sum_{j=1}^T \exp \left(\frac{X_j}{z} \right) \right)$ is the perspective function of the function *log-sum-exp* and according to [Boyd and Vandenberghe \(2004\)](#) the perspective function of a convex function is still a convex function. So the EVaR is the sum of a convex function and an affine function, so the EVaR is a convex function.

3 Entropic Optimization

3.1 Disciplined Convex Programming Framework of EVaR

First, we can express the perspective of the *log-sum-exp* as follow:

$$\begin{aligned}
t &\geq z \ln \left(\sum_{j=1}^T \exp \left(\frac{X_j}{z} \right) \right) \\
\frac{t}{z} &\geq \ln \left(\sum_{j=1}^T \exp \left(\frac{X_j}{z} \right) \right) \\
\frac{t}{z} - \frac{t}{z} &\geq \ln \left(\sum_{j=1}^T \exp \left(\frac{X_j}{z} \right) \right) - \ln \left(\exp \left(\frac{t}{z} \right) \right) \\
0 &\geq \ln \left(\sum_{j=1}^T \frac{\exp \left(\frac{X_j}{z} \right)}{\exp \left(\frac{t}{z} \right)} \right) \\
1 &\geq \sum_{j=1}^T \exp \left(\frac{X_j}{z} - \frac{t}{z} \right) \\
z &\geq \sum_{j=1}^T z \exp \left(\frac{X_j - t}{z} \right)
\end{aligned} \tag{10}$$

where t is a variable. If we use the variable u then:

$$\begin{aligned}
z &\geq \sum_{j=1}^T u_j \\
u_j &\geq z \exp \left(\frac{X_j - t}{z} \right) \quad \forall j = 1, \dots, T
\end{aligned} \tag{11}$$

We can express the minimization of the perspective of the function *log-sum-exp* as follows:

$$\min_z \quad z \log\text{-sum-exp}(X/z) = \begin{cases} \min_{z, t, u} & t \\ \text{s.t.} & z \geq \sum_{j=1}^T u_j \\ & (X_j - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \end{cases} \tag{12}$$

where K_{exp} is an exponential cone defined by [Chares \(2009\)](#) as:

$$K_{\text{exp}} = \left\{ (a, b, c) \mid b > 0, c \geq b \exp \left(\frac{a}{b} \right) \right\} \cup \left\{ (a, b, c) \mid a \leq 0, b = 0, c \geq 0 \right\} \tag{13}$$

The disciplined convex programming framework of EVaR is:

$$\text{EVaR}_\alpha(X) = \begin{cases} \min_{z, t, u} & t + z \ln \left(\frac{1}{\alpha T} \right) \\ \text{s.t.} & z \geq \sum_{j=1}^T u_j \\ & (X_j - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \end{cases} \quad (14)$$

3.2 EVaR Optimization

If $X = -rx^\tau$ and we replace the disciplined convex programming framework of EVaR in the five portfolio optimization problems from section 2, we obtain the following problems:

- Maximization of the expected return given a level of EVaR $\overline{\text{EVaR}}$.

$$\begin{aligned} \max_{x, z, t, u} \quad & \mu x^\tau \\ \text{s.t.} \quad & t + z \ln \left(\frac{1}{\alpha T} \right) \leq \overline{\text{EVaR}} \\ & \sum_{i=1}^N x_i = 1 \\ & z \geq \sum_{j=1}^T u_j \\ & (-r_j x^\tau - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\ & x_i \geq 0 ; \quad \forall i = 1, \dots, N \end{aligned} \quad (15)$$

where r_j is a vector of returns in period j for the N assets and x is the vector of weights of the assets.

- Minimization of the expected EVaR given a level of expected return $\bar{\mu}$.

$$\begin{aligned} \min_{x, z, t, u} \quad & t + z \ln \left(\frac{1}{\alpha T} \right) \\ \text{s.t.} \quad & \mu x^\tau \geq \bar{\mu} \\ & \sum_{i=1}^N x_i = 1 \\ & z \geq \sum_{j=1}^T u_j \\ & (-r_j x^\tau - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\ & x_i \geq 0 ; \quad \forall i = 1, \dots, N \end{aligned} \quad (16)$$

- Maximization of EVaR averse utility function given a risk aversion parameter $\lambda \in [0, \infty)$:

$$\begin{aligned}
& \max_{x, z, t, u} \quad \mu x^\tau - \lambda \left(t + z \ln \left(\frac{1}{\alpha T} \right) \right) \\
& \text{s.t.} \quad \sum_{i=1}^N x_i = 1 \\
& \quad \quad z \geq \sum_{j=1}^T u_j \\
& \quad \quad (-r_j x^\tau - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\
& \quad \quad x_i \geq 0 ; \quad \forall i = 1, \dots, N
\end{aligned} \tag{17}$$

- Maximization of the EVaR adjusted return assuming that the return of the risk free asset is r_f .

$$\begin{aligned}
& \min_{y, k, z, t, u} \quad t + z \ln \left(\frac{1}{\alpha T} \right) \\
& \text{s.t.} \quad \mu y^\tau - r_f k = 1 \\
& \quad \quad \sum_{i=1}^N y_i = k \\
& \quad \quad z \geq \sum_{j=1}^T u_j \\
& \quad \quad (-r_j y^\tau - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\
& \quad \quad k \geq 0 \\
& \quad \quad y_i \geq 0 ; \quad \forall i = 1, \dots, N
\end{aligned} \tag{18}$$

where the optimal portfolio is obtained making the transformation $x = y/k$.

- Risk parity EVaR portfolio.

$$\begin{aligned}
\min_{y, z, t, u} \quad & t + z \ln \left(\frac{1}{\alpha T} \right) \\
& \sum_{i=1}^N b_i \ln(y_i) \geq c \\
& \sum_{i=1}^N b_i = 1 \\
& z \geq \sum_{j=1}^T u_j \\
& (-r_j y^T - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\
& y_i \geq 0 ; \quad \forall i = 1, \dots, N
\end{aligned} \tag{19}$$

where the optimal portfolio is obtained making the transformation $x = y / \sum_{i=1}^N y_i$.

3.3 Entropic Drawdown at Risk (EDaR)

EDaR is an extension of EVaR applied to drawdowns distribution. The EDaR is the upper bound of drawdown at risk (DaR) and conditional drawdown at risk (CDaR) obtained from Chernoff inequality. Formally is defined as:

$$\begin{aligned}
\text{EDaR}_\alpha(X) &= \text{EVaR}_\alpha(\text{DD}(X)) \\
\text{EDaR}_\alpha(X) &= \inf_{z > 0} \left\{ z \ln \left(\frac{1}{\alpha} M_{\text{DD}(X)} \left(\frac{1}{z} \right) \right) \right\}
\end{aligned} \tag{20}$$

where $M_X(t)$ is the moment generating function of t , α is the significance level and $\text{DD}(X)$ is the drawdown of X .

3.4 EDaR Optimization

If we define the matrix of uncompound cumulative returns $R \in \mathbb{R}^{T \times N}$ where $R_k x^\tau = \sum_{i=1}^k r_i x^\tau$ and using the variable $d \in \mathbb{R}^{T+1 \times N}$ following the linear programming framework for drawdowns proposed by [Chekhlov et al. \(2000\)](#), we can express the five EDaR portfolio optimization problems as follows:

- Maximization of the expected return given a level of EDaR $\overline{\text{EDaR}}$.

$$\begin{aligned}
& \max_{x, z, t, u, d} \quad \mu x^\tau \\
& \text{s.t.} \quad t + z \ln \left(\frac{1}{\alpha T} \right) \leq \overline{\text{EDaR}} \\
& \quad \sum_{i=1}^N x_i = 1 \\
& \quad z \geq \sum_{j=1}^T u_j \\
& \quad (d_j - R_j x^\tau - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\
& \quad d_j \geq R_j x^\tau \quad \forall j = 1, \dots, T \\
& \quad d_j \geq d_{j-1} \quad \forall j = 1, \dots, T \\
& \quad d_j \geq 0 \quad \forall j = 1, \dots, T \\
& \quad d_0 = 0 \\
& \quad x_i \geq 0 ; \quad \forall i = 1, \dots, N
\end{aligned} \tag{21}$$

where R_j is a vector of uncompounded cumulative returns in period j for the N assets.

- Minimization of the expected EDaR given a level of expected return $\bar{\mu}$.

$$\begin{aligned}
& \min_{x, z, t, u, d} \quad t + z \ln \left(\frac{1}{\alpha T} \right) \\
& \text{s.t.} \quad \mu x^\tau \geq \bar{\mu} \\
& \quad \sum_{i=1}^N x_i = 1 \\
& \quad z \geq \sum_{j=1}^T u_j \\
& \quad (d_j - R_j x^\tau - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\
& \quad d_j \geq R_j x^\tau \quad \forall j = 1, \dots, T \\
& \quad d_j \geq d_{j-1} \quad \forall j = 1, \dots, T \\
& \quad d_j \geq 0 \quad \forall j = 1, \dots, T \\
& \quad d_0 = 0 \\
& \quad x_i \geq 0 ; \quad \forall i = 1, \dots, N
\end{aligned} \tag{22}$$

- Maximization of EDaR averse utility function given a risk aversion parameter

$\lambda \in [0, \infty)$.

$$\begin{aligned}
& \max_{x, z, t, u, d} \quad \mu x^\tau - \lambda \left(t + z \ln \left(\frac{1}{\alpha T} \right) \right) \\
& \text{s.t.} \quad \sum_{i=1}^N x_i = 1 \\
& \quad z \geq \sum_{j=1}^T u_j \\
& \quad (d_j - R_j x^\tau - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\
& \quad d_j \geq R_j x^\tau \quad \forall j = 1, \dots, T \\
& \quad d_j \geq d_{j-1} \quad \forall j = 1, \dots, T \\
& \quad d_j \geq 0 \quad \forall j = 1, \dots, T \\
& \quad d_0 = 0 \\
& \quad x_i \geq 0 ; \quad \forall i = 1, \dots, N
\end{aligned} \tag{23}$$

- Maximization of the EDaR adjusted return assuming that the return of the risk free asset is r_f .

$$\begin{aligned}
& \min_{y, k, z, t, u, d} \quad t + z \ln \left(\frac{1}{\alpha T} \right) \\
& \text{s.t.} \quad \mu y^\tau - r_f k = 1 \\
& \quad \sum_{i=1}^N y_i = k \\
& \quad z \geq \sum_{j=1}^T u_j \\
& \quad (d_j - R_j y^\tau - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\
& \quad d_j \geq R_j y^\tau \quad \forall j = 1, \dots, T \\
& \quad d_j \geq d_{j-1} \quad \forall j = 1, \dots, T \\
& \quad d_j \geq 0 \quad \forall j = 1, \dots, T \\
& \quad d_0 = 0 \\
& \quad k \geq 0 \\
& \quad y_i \geq 0 ; \quad \forall i = 1, \dots, N
\end{aligned} \tag{24}$$

where the optimal portfolio is obtained making the transformation $x = y/k$.

- Risk parity EDaR portfolio.

$$\begin{aligned}
& \min_{y, z, t, u, d} \quad t + z \ln \left(\frac{1}{\alpha T} \right) \\
& \sum_{i=1}^N b_i \ln(y_i) \geq c \\
& \sum_{i=1}^N b_i = 1 \\
& z \geq \sum_{j=1}^T u_j \\
& (d_j - R_j y^T - t, z, u_j) \in K_{\text{exp}} \quad \forall j = 1, \dots, T \\
& d_j \geq R_j y^T \quad \forall j = 1, \dots, T \\
& d_j \geq d_{j-1} \quad \forall j = 1, \dots, T \\
& d_j = 1 \quad \geq \quad 0 \quad \forall \quad j = 1, \dots, T \\
& d_0 = 0 \\
& y_i \geq 0 ; \quad \forall i = 1, \dots, N
\end{aligned} \tag{25}$$

where the optimal portfolio is obtained making the transformation $x = y / \sum_{i=1}^N y_i$.

4 Numerical Examples

We select 30 assets (i.e., stocks JCI, TGT, CMCSA, CPB, MO, T, APA, MMC, JPM, ZION, PSA, BAX, BMY, AAPL, PCAR, BA, TMO, TXT, DE, MSFT, HPQ, SEE, VZ, CNP, NI, JNJ, PFE, AMZN, GE and GOOG) from the S&P 500 (NYSE) and download daily adjusted closed prices from Yahoo Finance for the period from January 1, 2016 to December 30, 2019. Then, we calculated daily returns building a returns matrix of size $T = 1004$ and $N = 30$. To calculate the portfolios we consider a significance level of 5% and we use Python 3.7², Riskfolio-Lib³ and MOSEK⁴ solver because it supports exponential cone programming.

4.1 EVaR Numerical Example

In this section, we calculate the portfolio that maximize the EVaR adjusted return, the EVaR efficient frontier and risk parity EVaR portfolio.

²<https://winpython.github.io>

³<https://github.com/dcajasn/Riskfolio-Lib>

⁴<https://www.mosek.com>

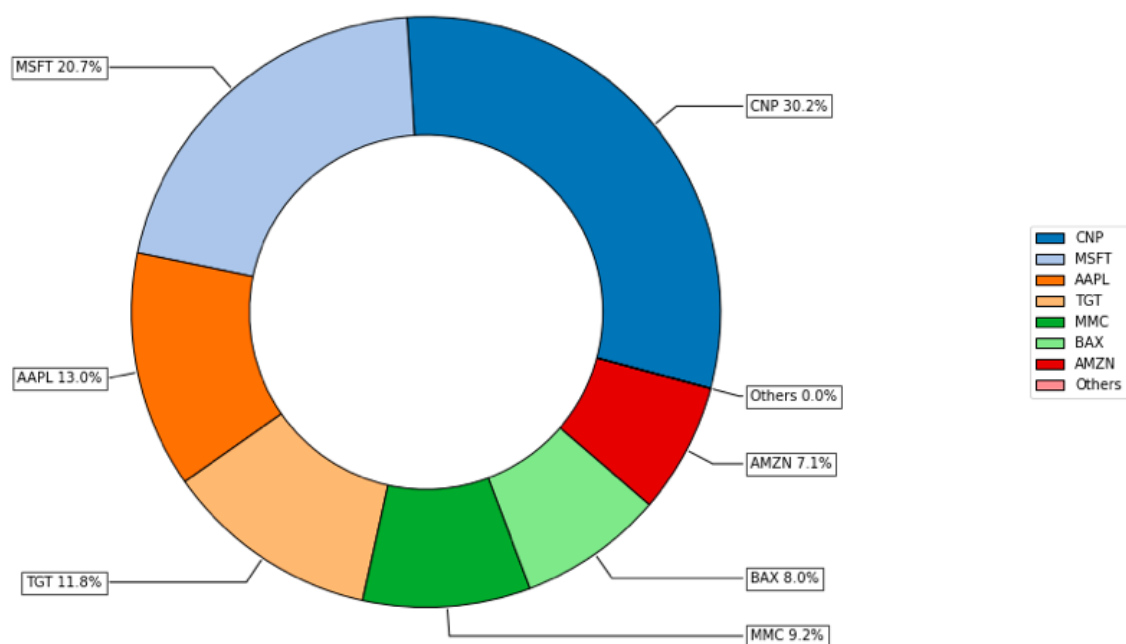


Figure 1: Composition Max EVaR Adjusted Return Portfolio

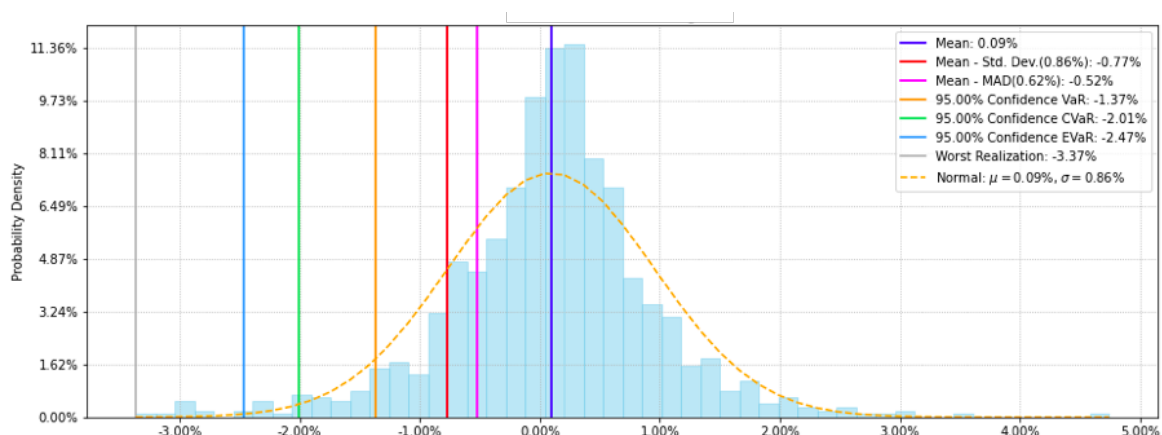


Figure 2: Histogram Max EVaR Adjusted Return Portfolio

If we see the histogram of max EDaR adjusted return portfolio in Figure 2, the VaR is 1.26%, CVaR is 2.09% and EVaR is 2.62%. The EVaR is more than twice times the VaR and 0.53% higher than CVaR, this is an advantage for model risk because with the same confidence level we have a higher bound for losses (capital requirements) without the need to increase confidence level.

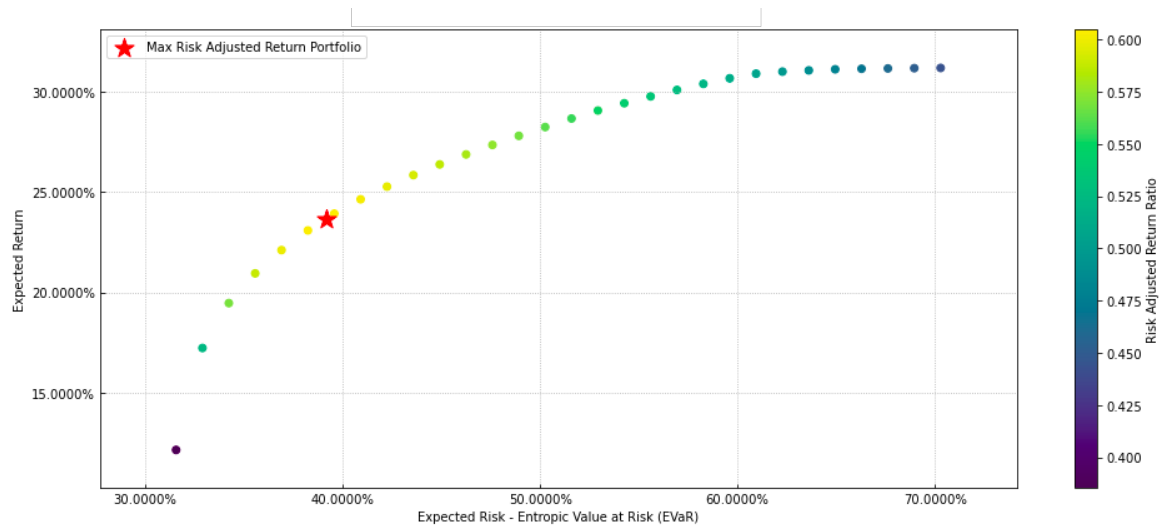


Figure 3: Efficient Frontier Return-EVaR

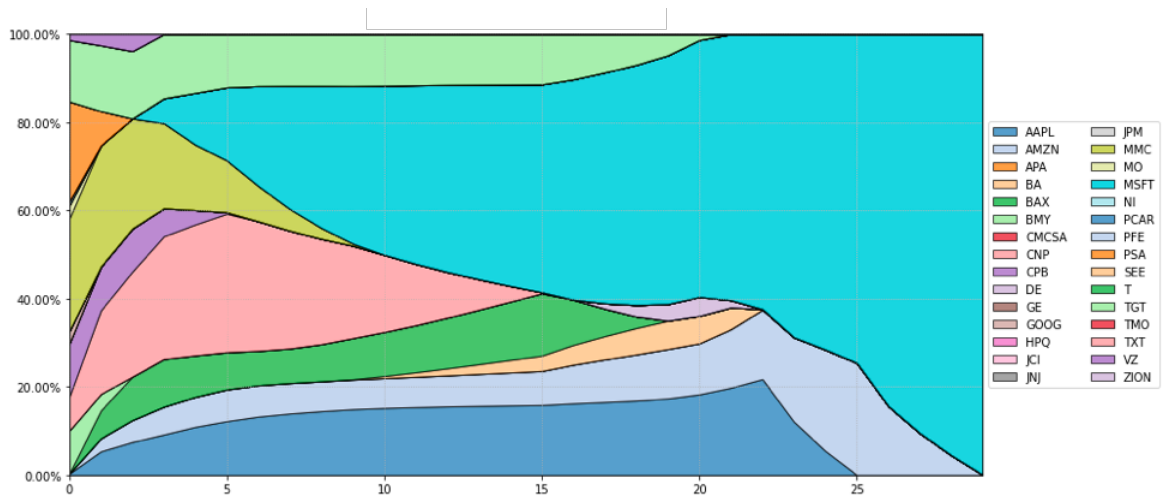


Figure 4: Composition Efficient Frontier Return-EVaR

To build the efficient frontier Return-EVaR we solve the problem maximize the portfolio return subject to a constraint on EVaR for several values of EVaR.

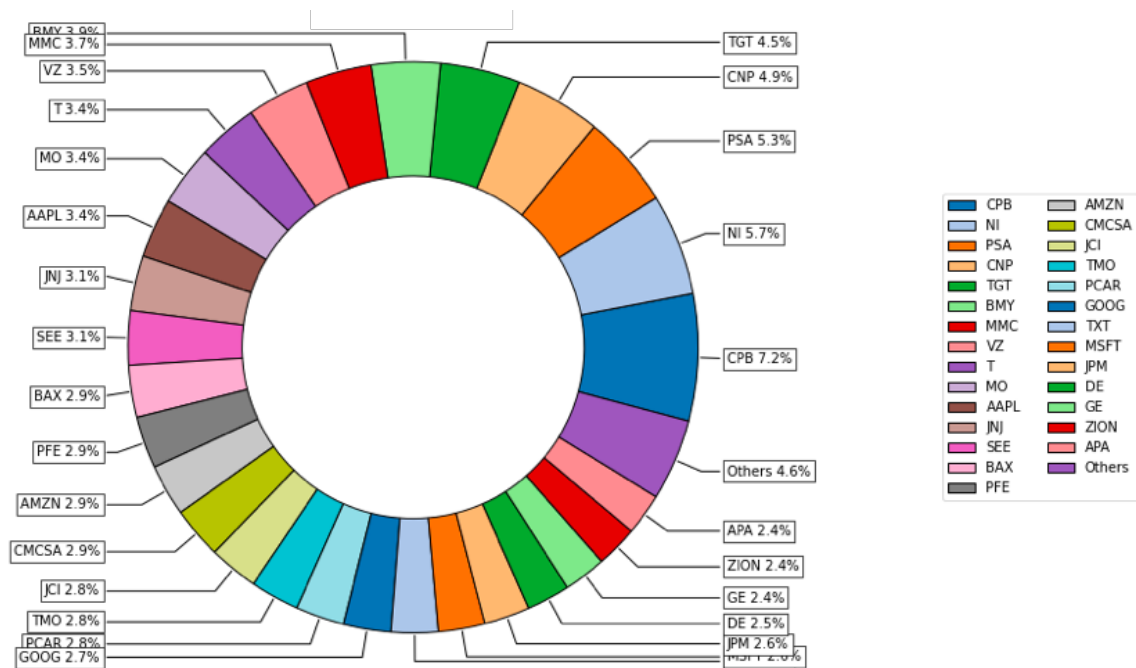


Figure 5: Composition Risk Parity EVaR Portfolio

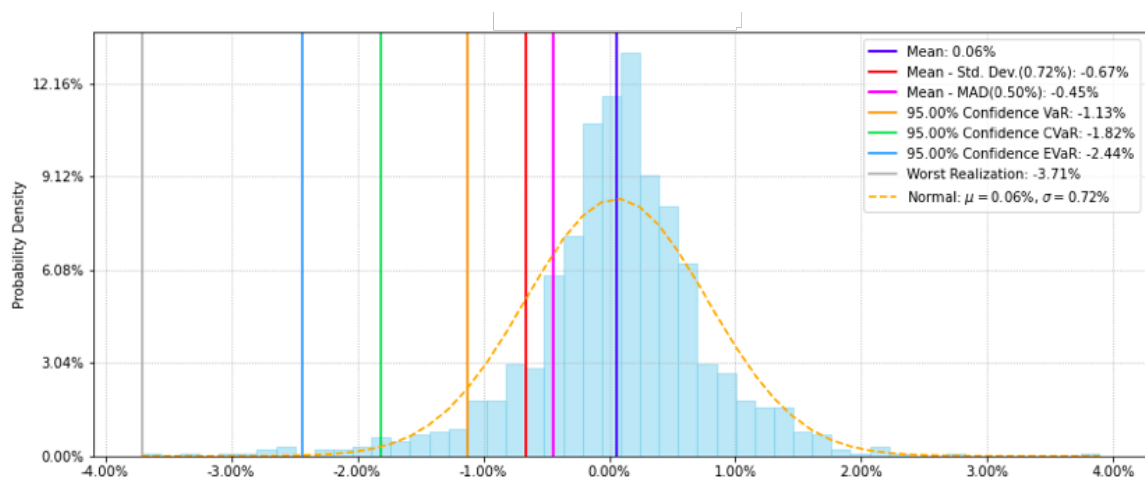


Figure 6: Histogram Risk Parity EVaR Portfolio

In the histogram of risk parity EVaR portfolio in Figure 6, the VaR is 1.18%, CVaR is 1.84% and EVaR is 2.46%. The EVaR is more than twice times the VaR and 0.62% higher than CVaR.

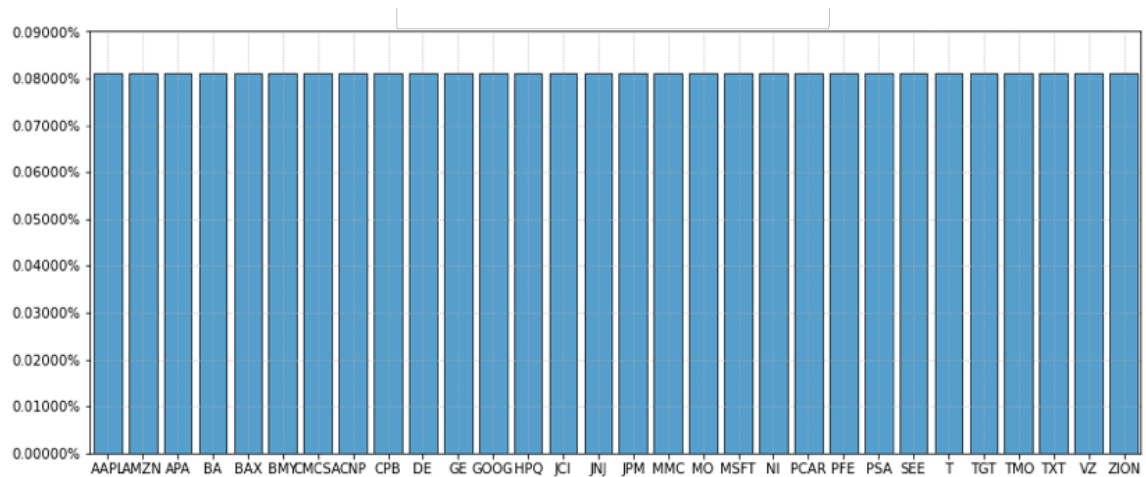


Figure 7: EVaR Contribution per Asset

In Figure 7, we can see that risk parity EVaR portfolio effectively breaks down EVaR evenly across all assets. This shows that the solution of risk parity EVaR problem that we posed fulfills its objective.

4.2 EDaR Numerical Example

In this section, we calculate the portfolio that maximize the EDaR adjusted return, the EDaR efficient frontier and risk parity EDaR portfolio.

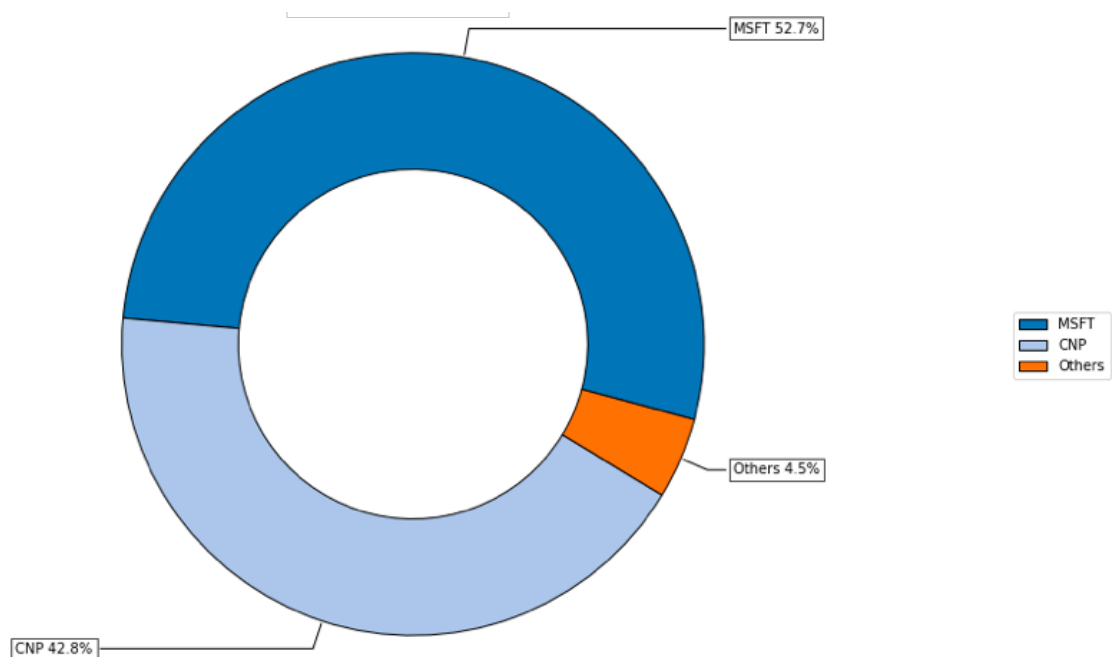


Figure 8: Composition Max EDaR Adjusted Return Portfolio

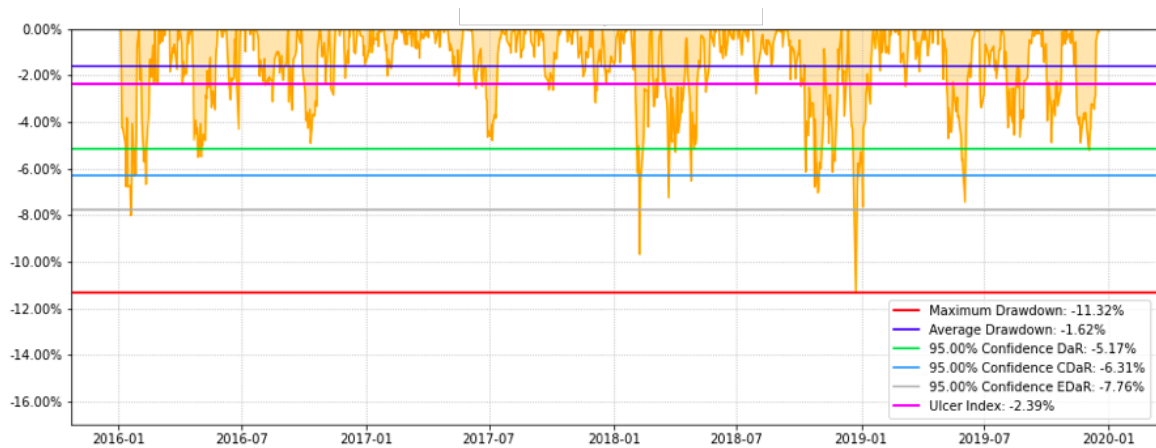


Figure 9: Uncompounded Cumulative Drawdowns Max EDaR Adjusted Return Portfolio

If we see the drawdown plot of max EDaR adjusted return portfolio in Figure 9, the drawdown at risk (DaR) is 5.17%, conditional drawdown at risk (CDaR) is 6.31% and EDaR is 7.76%. The EDaR is higher 2.59% than VaR and 1.45% higher than CDaR, this is an advantage for model risk because with the same confidence level we have a higher bound for drawdowns without the need to increase confidence level.

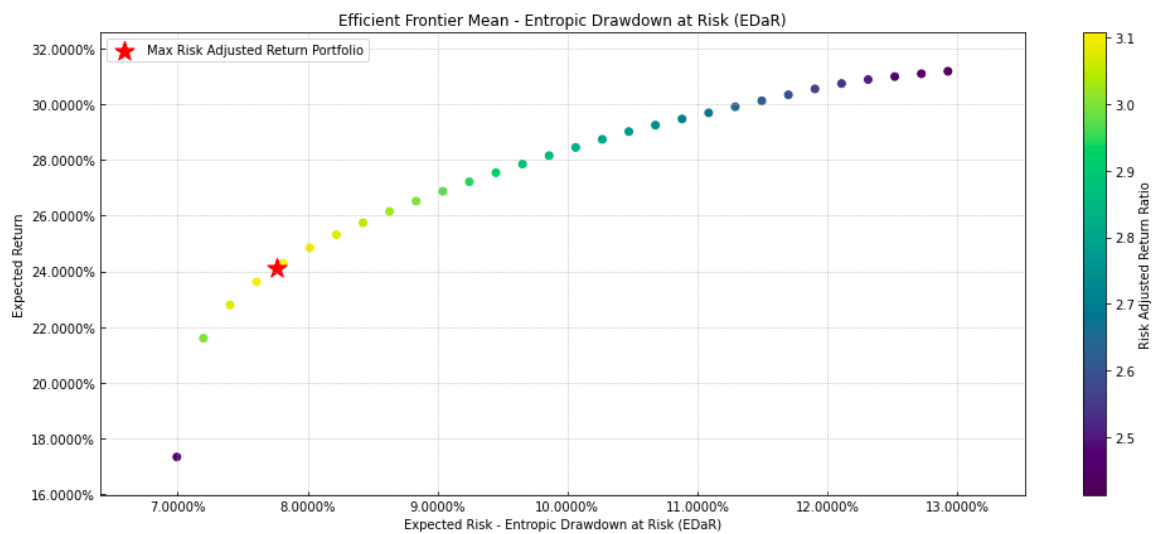


Figure 10: Efficient Frontier Return-EDaR

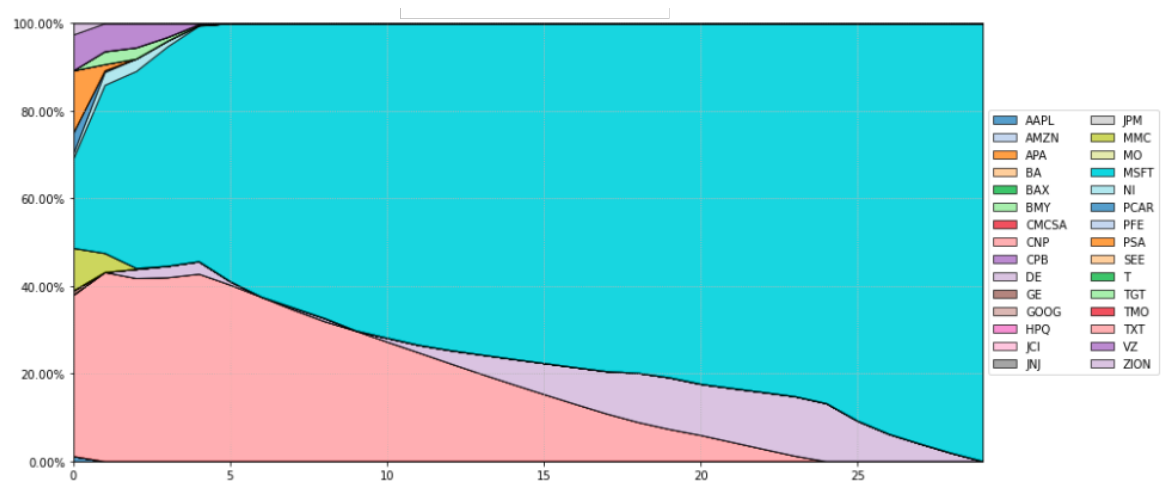


Figure 11: Composition Efficient Frontier Return-EDaR

To build the efficient frontier Return-EDaR we solve the problem maximize the portfolio return subject to a constraint on EDaR for several values of EDaR.

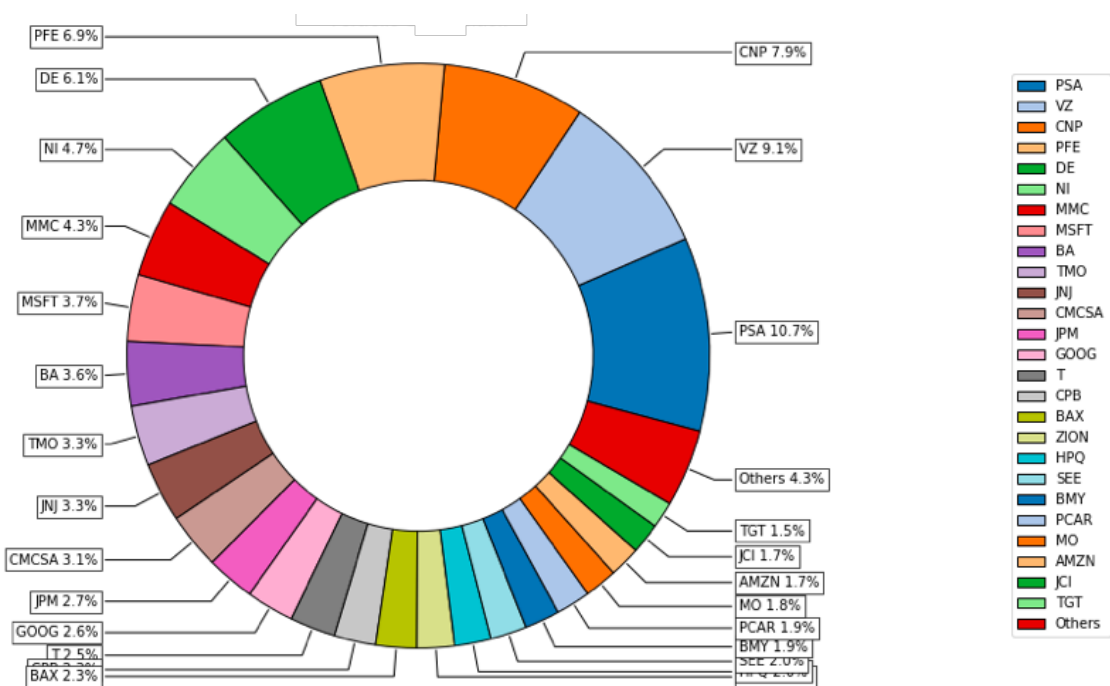


Figure 12: Composition Risk Parity EDaR Portfolio

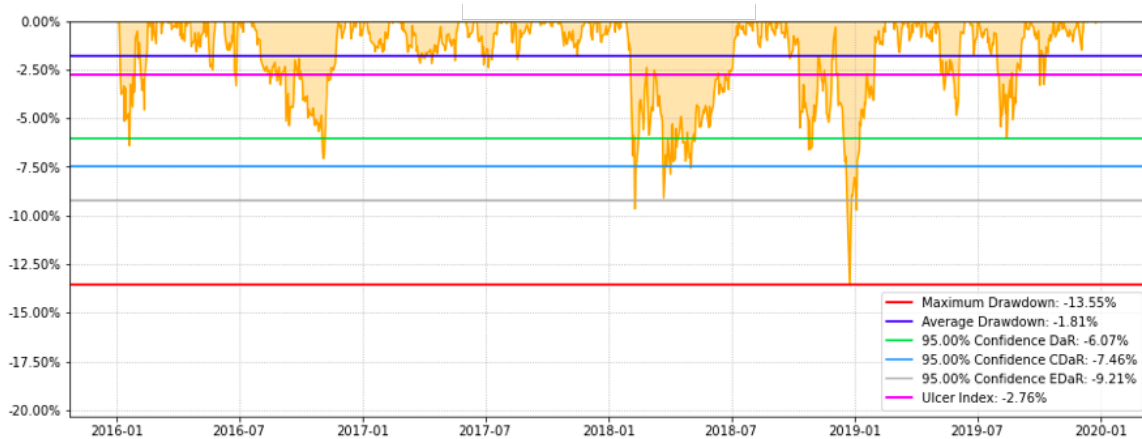


Figure 13: Uncompounded Cumulated Drawdowns Risk Parity EDaR Portfolio

In the drawdown plot of risk parity EDaR portfolio in Figure 13, the DaR is 6.07%, CDaR is 7.46% and EDaR is 9.21%. The EDaR is higher 3.14% than DaR and 1.75% higher than CDaR.

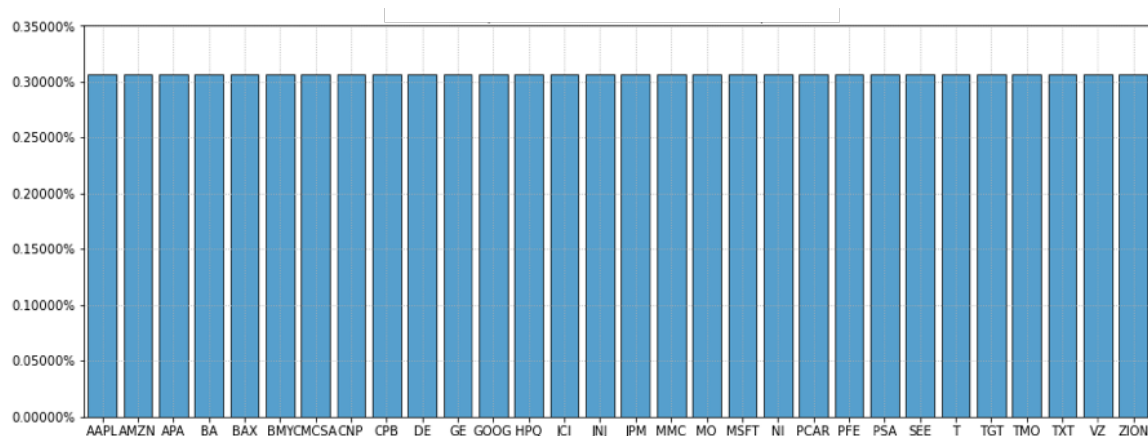


Figure 14: EDaR Contribution per Asset

In Figure 14, we can see that risk parity EdaR portfolio effectively breaks down EDaR evenly across all assets. This shows that the solution of risk parity EDaR problem that we posed fulfills its objective.

5 EVaR and EDaR in Large Scale Problems

In this section we compare the performance of EVaR and EDaR models respect to CVaR and CDaR formulations proposed by Rockafellar and Uryasev (2000) and Chekhlov et al. (2000) respectively. We simulated 50,000 normal returns for 10, 50, 100 and 200

assets. Then, we estimated 5 instances of the portfolios that maximizes the risk adjusted return and minimize risk for each risk measure and sample size. Then, we computed the average time for each combination of sample size, objective function and risk measure. The simulation was computed using a computer with four 3.6-GHz processors and 8 GB of RAM, Python 3.7, Riskfolio-Lib package and MOSEK solver.

Objective	Assets	CVaR	Δ	EVaR	Δ
Min Risk	10	3.4		29.5	
	50	8.0	4.6	47.4	17.9
	100	15.4	7.4	61.7	14.3
	200	31.5	16.1	85.7	24
Return/Risk	10	2.5		36.1	
	50	7.5	5.0	61.6	25.5
	100	14.8	7.3	67.5	5.9
	200	31.9	17.1	91.6	24.1

Table 1: Average run time in seconds of CVaR and EVaR portfolios

We can see in Table 1 that average run time of EVaR portfolios are higher than CVaR portfolios due to exponential cone constraints are harder to solve than linear constraints. On the other hand, the results are near to the obtained by [Ahmadi-Javid and Fallah-Tafti \(2019\)](#), the differences are 36.4, 36.6 and 12.7 in the cases of 50, 100 and 200 assets respectively.

Objective	Assets	CDaR	Δ	EDaR	Δ
Min Risk	10	17.4		75.1	
	50	40.2	22.8	107.2	32.1
	100	74.5	34.3	160.6	53.4
	200	144.8	70.3	225.9	65.3
Return/Risk	10	19.4		126.9	
	50	52.4	33	185	58.1
	100	97	44.6	281.2	96.2
	200	162.5	65.5	640.1	358.9

Table 2: Average run time in seconds of CDaR and EDaR portfolios

We can see in Table 2 that average run time of EDaR portfolios are higher than CDaR portfolios and that the run time increases more in EDaR than CDaR when we increase the number of assets.

6 Conclusions

EVaR is a more conservative risk measure that allows us to get higher losses bounds without the need to reduce the significance level. This is a good feature for model risk because we do not have the need to increase confidence level to get a more conservative bound for losses.

This work presents a disciplined convex programming framework of EVaR based on exponential cone programming, that allows us to propose five EVaR portfolio optimization problems. Also, we applied the concept of EVaR to drawdowns distribution and developed five EDaR portfolio optimization problems. Then, we make some examples using Python, Riskfolio-Lib package and MOSEK solver. We built an efficient frontier, a maximum risk adjusted return portfolio, and a risk parity portfolio for EVaR and EDaR. Also, we performed a simulation to test the efficiency of the proposed frameworks and we observed that the new EVaR framework has similar performance than the algorithm proposed by [Ahmadi-Javid and Fallah-Tafti \(2019\)](#), but average run time are higher than CVaR. In the case of new EDaR framework, we observed that average run time respect to CDaR is higher.

The models proposed in this work allows users to use disciplined convex programming softwares that supports exponential cone programming for EVaR and EDaR optimization. This makes portfolio optimization based on EVaR and EDaR accessible to most users.

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